Empirical rule with z-scores

1 -2

2 3

3

a $x = 65\%$

b $x = 68\%$

4

a 0.13

b 147

5 Mathematics

6 English

7

a 2.7

b 3.7

c Gwen

8

a -1.2

b -2.9

c Mathematics

9

a 0.9265 units^2

b 0.0570 units^2

c 0.1093 units^2

d 0.8790 units^2

e 0.0336 units^2

f 0.0809 units^2

10

a 93.45%

b 86.40%

11

a 0.1593

b 0.6607

c 0.9274

d 0.8527

Calculations with the standard and general normal distributions

12

a 0.8528

b -0.8183

c -0.7790

d 1.0102

e 0.8638

f 1.0141

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13
a 21.93 **b** 20.63 **c** 10.54 **d** 30.19

14
a 3 **b** 91 **c** 2.4 **d** 76.8

Quantiles and percentiles

15
a 0.524 **b** 0.385 **c** 0.842

16
a 0 **b** 1 **c** -1

17
a 45 **b** 40 **c** 60

18
a 41 cm **b** 37 cm

19
a 91.584 **b** 112.816 **c** 94.756

20
a 30 **b** 33.82 **c** 28.46

21
a 1.600 **b** -0.385 **c** 48.075

22 To be accepted into Mensa, a person must score higher than 98% of other people on the IQ test or a successful applicant's IQ score will be among the top 2% of the population.

Unknown mean or standard deviation

23 46

24 60

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26 6.5

27 11.54

28 16.02

29 $\mu = 73.33, \sigma = 3.67$

30 $\mu = 14.02, \sigma = 3.57$

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